A Quartic in Ninety-One Variables under the Rational *abc* Conjecture

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Abstract

Assuming the rational *abc* conjecture and the specified elliptic-group, pole-realization and simultaneous prime-pattern inputs, we construct a family of integer quartics in ninety-one total variables. Every sufficiently large fixed odd input shift gives rational input projection exactly $\mathbb{Z}$. The quartic is a sum of seventy-nine primitive quadratic squares and has exactly 926 monomials. Its coefficient height, for a shift $B \geq 3$, is $\max(4153456, B^2 + 797683)$. A common elliptic chart in four field coordinates stores both reciprocal formal parameters. Eight rational contacts symmetric about one permit a short quadratic evaluation and normalization. A height estimate for a sum of fourth powers bounds the input together with one elliptic index. All six local tests have order one; only plain elliptic formal parameters are used. Reciprocal factor equations avoid separately stored numerator fourth powers, and their zero stratum is excluded by the point-pair equations. The sufficient shift is ineffective.

1 Statement and arithmetic assumptions

For a reduced fraction u/v, with $v > 0$, put $h(u/v) = \log \max(|u|, v)$. Write $h(1 : u : v)$ for the logarithmic projective height of the indicated rational point. The coefficient height of an integer polynomial is the maximum absolute value of its coefficients. All variable counts include the distinguished input.

Theorem 1.1. *Under the assumptions below, there is a computable family*

$$P_B \in \mathbb{Z}[X, Y_1, \ldots, Y_{90}], \qquad B \geq 3 \text{ odd},$$

such that every member has degree exactly four, content one, exactly 926 *monomials, and coefficient height*

$$H(P_B) = \max(4153456, B^2 + 797683).$$

There is a constant B_* such that, for every odd $B \geq B_*$,

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{90}, \quad P_B(x, \boldsymbol{y}) = 0.$$

The integer B is a fixed coefficient parameter. No numerical sufficient value of B_* is determined. In particular, the coefficient fixture $B = 3$ is not asserted to define the integers.

Let $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, let σ be conjugation, and put

$$d = 3 - \rho, \quad D = d^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = (x_P, y_P) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

The arithmetic assumptions have the following precise scope.

1. The rational *abc* conjecture: for every $\epsilon > 0$ there is $K_\epsilon \geq 1$ such that $c \leq K_\epsilon \operatorname{rad}(abc)^{1+\epsilon}$ for pairwise coprime positive integers $a + b = c$.

2. The elliptic source gives $2E(K) = \mathbb{Z}P$, with P of infinite order, and the index-height comparison stated in [2]. The square-abscissa characterization proved in [4] then applies to all nonzero multiples of P.

3. For a nonzero rational b, set $B_0 = 17b$. The pole equations

$$\begin{aligned} p_0^2 &= B_0(1 - e_0)(1 + 3e_0), \\ B_0 q_0^2 &= (s_0^2 - B_0 e_0(B_0 - b^2))(s_0^2 + 3B_0 e_0(B_0 - b^2)) \end{aligned} \tag{1}$$

   exclude negative odd valuations of b outside $\{2, 3, 17\}$. For inputs satisfying

$$\begin{gathered} 0 < b < 17, \qquad b \text{ has square reduced denominator}, \\ b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2, \end{gathered} \tag{2}$$

   the retained restricted-converse and descent hypotheses give a witness with $p_0 q_0 s_0 \neq 0$. The hypotheses and their nonzero-witness consequence are specified in [2, 7, 8].

4. The simultaneous prime-pattern input has exactly the quantified form in [2, 3]: arbitrary fixed rational contact coefficients after denominator clearing, the prescribed Gaussian congruences and conjugate reduction orders, fixed avoidance and divisibility, and compatible real approximation. Its analytic source is Sun's specified Theorem 4.1 and the estimates pinned in those papers. All scales and finite preparation data are fixed before prime selection.

The exponent-nineteen contact-height input is not used here.

The one-point case of the proved aggregate trace-height theorem [5], obtained from von Buhren's family height theorem, gives constants $A > 0$, $J \geq 0$ such that

$$n^2 \leq A h(T(n)) + J, \qquad T(n) = \operatorname{Tr} x([n]P), \quad n \in \mathbb{Z} \setminus \{0\}. \tag{3}$$

The additive constant is ineffective. Each abscissa is a positive square at both real embeddings and exceeds d there, so $T(n) > 6$.

2 Two height consequences of *abc*

Set

$$\Theta_8 = \{1 \pm 1, 1 \pm 2, 1 \pm 3, 1 \pm 4\} = \{-3, -2, -1, 0, 2, 3, 4, 5\}.$$

For $t \notin \Theta_8$, let $M(t)$ be the product of primes $p > 76176$ for which $v_p(t - \theta) > 0$ is odd for some $\theta \in \Theta_8$.

Lemma 2.1. *Assume rational abc. There is a fixed $C_0 \geq 0$ such that*

$$h(t) \leq \frac{1}{3} \log M(t) + C_0.$$

Proof. For a primitive pair (u, v), $v > 0$, use the squarefree binary form $G(u, v) = \prod_{\theta \in \Theta_8}(u - \theta v)$ of degree eight. Granville's consequence of *abc* [1, Proposition 1], with $\epsilon = 1/2$, gives

$$\log \operatorname{rad}(G(u, v)) \geq \frac{11}{2} \log \max(|u|, v) - C.$$

For a nonzero integer m, let $\operatorname{sf}(m)$ be the product of primes having odd valuation in m. Prime by prime, $\operatorname{rad}(m)^2 \leq |m| \operatorname{sf}(m)$. Also $|G(u, v)| \leq C' \max(|u|, v)^8$. Outside the fixed cutoff, no

prime divides two distinct factors of G, and a prime dividing v divides none of them. Its squarefree part outside the cutoff is therefore precisely $M(u/v)$. The omitted squarefree factors are bounded by a fixed product of primes. Consequently

$$\log M(u/v) \geq \left(2 \cdot \frac{11}{2} - 8\right) h(u/v) - C'' = 3h(u/v) - C'',$$

as required. □

Lemma 2.2. *Assume rational abc. There is a fixed $C_1 \geq 0$ such that*

$$h(1 : u : v) \leq \frac{4}{3} h(u^4 + v^4) + C_1 \qquad (u, v \in \mathbb{Q}).$$

One may take $C_1 = \frac{7}{12} \log 2 + \frac{5}{4} \log K_{1/15}$.

Proof. The zero-coordinate cases follow from $h(w^4) = 4h(w)$. Otherwise write $u = gA/D_0$, $v = gB/D_0$, where $\gcd(A, B) = \gcd(g, D_0) = 1$, $g, D_0 > 0$. Put $H = \max(|A|, |B|)$ and $N = A^4 + B^4$, so $H^4 \leq N \leq 2H^4$. Let $k = \gcd(N, D_0^4)$, $m_0 = N/k$, $n_0 = D_0^4/k$. Then $u^4 + v^4 = g^4 m_0/n_0$ is reduced. Write $M = \max(g^4 m_0, n_0)$.

Every prime dividing N either divides D_0 or survives in m_0, so $\mathrm{rad}(N) \leq D_0 m_0$. Apply *abc* with $\epsilon = 1/15$ to $A^4 + B^4 = N$. The three positive terms are pairwise coprime, and

$$H^{7/4} \leq K_{1/15}^{15/16} D_0 m_0.$$

Using $D_0^4 = n_0 N/m_0 \leq 2n_0 H^4/m_0$ gives

$$H^{3/4} \leq 2^{1/4} K_{1/15}^{15/16} m_0^{3/4} n_0^{1/4}.$$

Moreover $\max(gH, D_0) \leq 2^{1/4} H (M/m_0)^{1/4}$. Taking logarithms and combining yields

$$\begin{aligned} h(1 : u : v) &= \log \max(gH, D_0) \\ &\leq C_1 + \frac{3}{4} \log m_0 + \frac{1}{3} \log n_0 + \frac{1}{4} \log M \leq C_1 + \frac{4}{3} \log M. \end{aligned}$$

This includes common numerator factors and all denominator cancellation. □

3 Quadratic point equations

Put $\beta = 2X^2 + B$. If $X = a/b$ is reduced, integrality of β implies $b^2 \mid 2$, hence $b = 1$. Thus $\beta \in \mathbb{Z}$ if and only if $X \in \mathbb{Z}$; for integer input β is odd, and always $\beta \geq B$.

All equations in this section are over K and mean their two rational component equations. Products of stored quantities have degree at most two. For the root point retain $h, j, a \in K$ and impose

$$a = h^2, \qquad j^2 = a^2 - D. \tag{4}$$

Then $A = (a, hj) = [n]P$ for some nonzero integer n by the square-abscissa theorem [4].

3.1 The common reference point

Write $a_P = x_P$, $b_P = y_P$, and set

$$A_P = a_P^2 + D, \qquad B_P = a_P^2 - D, \qquad C_P = D^2 + 6a_P^2 D - 3a_P^4.$$

The constants a_P, b_P, D, A_P, C_P are nonzero. The first four nonvanishing assertions follow from the displayed coordinates and $D = (3-\rho)^2 > 0$ at both real embeddings; exact arithmetic gives

$$C_P = -\frac{94897379}{4096} + \frac{4195821}{256}\rho, \qquad \mathrm{N}(C_P) = -\frac{8203357319351}{16777216} \neq 0.$$

Retain four field coordinates S, T_H, f, F_0. Put

$$\begin{aligned} L_f &= 2b_P D(a_P T_H - b_P), \\ Q_f &= a_P A_P S T_H + b_P A_P S - b_P B_P T_H + 4a_P^2 D, \\ L_g &= 2b_P D(a_P S + b_P), \\ Q_g &= -a_P A_P S T_H - b_P B_P S + b_P A_P T_H - 4a_P^2 D. \end{aligned}$$

Impose the three quadratic equations

$$\begin{aligned} L_f f &= Q_f, \\ L_g F_0 &= Q_g, \\ D^2 f F_0 + a_P D(f + F_0) + a_P S T_H + b_P(S - T_H) + D &= 0. \end{aligned} \tag{5}$$

Lemma 3.1. *These equations describe exactly the points $H \in E(K)$ for which H and $H - P$ are finite with nonzero abscissas, through*

$$S = \frac{y(H)}{x(H)}, \qquad T_H = \frac{y(H-P)}{x(H-P)}, \qquad f = \frac{1}{x(H)}, \qquad F_0 = \frac{1}{x(H-P)}.$$

In particular $fF_0 \neq 0$ at every solution. The statement includes all finite exceptional cases of the displayed equations.

Proof. For the algebraic calculation abbreviate $a = a_P$, $b = b_P$, $T = T_H$ and $g = F_0$. Thus $b^2 = a(a^2 - D)$. Define

$$\Phi = aS^2T^2 - (a^2 - D)(S^2 + T^2) + 2(a^2 + D)ST + 4aD$$

and the polynomial expressions

$$\begin{aligned} 2bDf_0 &= -(a^2 + D)S + (a^2 - D)T - bS^2 - aS^2T, \\ 2bDg_0 &= (a^2 + D)T - (a^2 - D)S - bT^2 + aST^2. \end{aligned}$$

Direct expansion gives

$$\begin{aligned} Df_0^2 + S^2 f_0 - 1 &= \frac{a^2S^2 - b^2}{4ab^2D}\Phi, \\ Dg_0^2 + T^2 g_0 - 1 &= \frac{a^2T^2 - b^2}{4ab^2D}\Phi, \\ L_f f_0 - Q_f = -a\Phi, \qquad & L_g g_0 - Q_g = a\Phi. \end{aligned} \tag{6}$$

Let $r_f = L_f f - Q_f$, $r_g = L_g g - Q_g$, and $R = D^2 fg + aD(f + g) + aST + b(S - T) + D$. The further exact identity

$$\begin{aligned} L_f L_g R = D^2 r_f r_g &+ (aDL_g + D^2 Q_g) r_f \\ &+ (aDL_f + D^2 Q_f) r_g - aD^2 C_P \Phi \end{aligned} \tag{7}$$

shows that every solution has $\Phi = 0$, including the cases in which one leading coefficient vanishes. Also

$$(2bD)^2 R(f_0, g_0) = D^2\big((a^2 - D) - aST - b(S - T)\big)\Phi.$$

If $L_f L_g \neq 0$, these identities give $f = f_0, g = g_0$. If $L_f = 0$, then $T = b/a$ and $\Phi = 0$ forces

$$S = \frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad L_g = -\frac{aDC_P}{a^2 + D} \neq 0.$$

Thus $g = g_0 = 1/a$. The coefficient of f in R is $D(a^2 + D)/a \neq 0$, so $f = f_0$. The case $L_g = 0$ is symmetric. In particular both leading coefficients cannot vanish simultaneously; at $S = -b/a, T = b/a$ one has $Q_f = C_P \neq 0$.

Equation (6) now proves $fg \neq 0$ and gives the points $H = (f^{-1}, S/f)$ and $H' = (g^{-1}, T/g)$ on E. Two more identities prove the translation:

$$a(f_0 T + g_0 S) + b(g_0 - f_0) = S + T,$$

$$f_0 g_0 (aS + b)^2 - a(1 - af_0)^2 = -\frac{(aS + b)^2 (aST + b(S - T) + D)}{4b^2 D^2} \Phi.$$

If $x(H) \neq a$, the second identity gives the abscissa of $H - P$ from the chord slope $(x(H)S + b)/(x(H) - a)$. The first states that $H, -H', -P$ are collinear, and therefore gives the required ordinate. If $x(H) = a$, then $S = \pm b/a$. At $S = b/a$ the formula gives $f = -a/D \neq 1/a$, which is impossible. At $S = -b/a$ it gives $H = -P$ and

$$T = -\frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad g = \frac{4b^2}{(a^2 + D)^2},$$

which are the coordinates of $H' = -2P$ in the stated form. Thus $H' = H - P$ in all cases.

Conversely, the chord-and-tangent formulas for any allowed H give $\Phi = 0$ and the expressions f_0, g_0 above; the same explicit doubling computation covers $H = -P$. The displayed identities then prove all three equations. This proves both directions. □

Write $H = [m]R_0 + T_0$, with $T_0 \in E[2](K)$, and put $h_* = m/2$. This notation is an elliptic index; it does not define multiplication of a point by a nonintegral rational number.

3.2 A pair of points with fixed difference

Retain $V, Q, W_A, I_A, K_A \in K$ satisfying

$$\begin{aligned} hI_A &= (j + DW_A)V + aQ, \\ hK_A &= -(j + DW_A)Q - aV, \\ VQ &= aW_A, \\ V^2 + Q^2 + 2jW_A + DW_A^2 &= 1. \end{aligned} \tag{8}$$

Lemma 3.2. *Equations (4) and (8) describe points $U, V_1 \in \mathbb{Z}P$ with $U - V_1 = A$, allowing either numerator point to be the identity. Both I_A and K_A are nonzero. On the nonidentity branches,*

$$U = \left(\frac{a}{V^2}, \frac{aI_A}{V^3}\right), \qquad V_1 = \left(\frac{a}{Q^2}, \frac{aK_A}{Q^3}\right), \qquad z(U) = -\frac{V}{I_A}, \quad z(V_1) = -\frac{Q}{K_A},$$

where $z = -x/y$. In all branches the quantities $R_U, R_1 \in K$ defined by

$$I_A^2 = a - DaR_U, \qquad K_A^2 = a - DaR_1 \tag{9}$$

are $f(U)^2, f(V_1)^2$, with $f = 1/x$ extended by zero at the identity.

Proof. The square chart gives $haj \neq 0$. Squaring the first two equations and using the last two gives $aI_A^2 = a^2 - DV^4$ and $aK_A^2 = a^2 - DQ^4$. For $VQ \neq 0$ these are exactly the two curve equations. Their abscissas are squares, so the points lie in $\mathbb{Z}P$. The line with slope $(1/W_A - j)/h$ through U also passes through $-V_1$ and $-A$. Substitution gives its three abscissas $a/V^2, a/Q^2, a$, counted with multiplicity. The chord-and-tangent law therefore gives $U - V_1 = A$, including the tangent case. Nontorsion of the nonidentity points implies $I_A K_A \neq 0$.

If $V = 0$, then $W_A = 0$, $Q^2 = 1$, $I_A = hQ$ and $K_A = -jQ/h$. The interpretation is $U = O, V_1 = -A$. If $Q = 0$, then $W_A = 0$, $V^2 = 1$, $I_A = jV/h$ and $K_A = -hV$, giving $U = A, V_1 = O$. The two zero cases cannot occur together. The asserted nonvanishing and reciprocal-square identities also hold there.

Conversely, let $U, V_1 \in \mathbb{Z}P$ be nonidentity points with $U - V_1 = A$. Their abscissas are squares. The line through $U, -V_1, -A$ has equation $y = \mu x + c$, with $c = -h(\mu h + j)$. Put $R = \mu h + j$. Vieta's identities give $x(U)x(V_1) = R^2$ and $x(U) + x(V_1) = \mu^2 - a$; in particular $R \neq 0$. Choose square roots $u_0^2 = x(U)$, $v_0^2 = x(V_1)$ with $u_0 v_0 = R$, and put

$$V = h/u_0, \qquad Q = h/v_0, \qquad W_A = 1/R, \qquad I_A = y(U)V^3/a, \qquad K_A = y(V_1)Q^3/a.$$

The two Vieta identities give the last two chart equations, and the line ordinates give the first two. This also covers tangent lines. The identity branches are supplied by the explicit formulas above. □

3.3 The two formal-parameter ratios

Retain $L, L_0 \in K$ and impose the quadratic equations

$$I_A L = VS, \qquad K_A L_0 = QT_H. \tag{10}$$

The pair-chart lemma gives $I_A K_A \neq 0$. Once H and $H - P$ are nontorsion, the common-point lemma and $z = -x/y$ identify these values as

$$L = \frac{z(U)}{z(H)}, \qquad L' = \sigma L_0 = \sigma\left(\frac{z(V_1)}{z(H - P)}\right).$$

The equations also apply when U or V_1 is the identity, with its formal parameter interpreted as zero. No additional coordinate for either numerator parameter is required.

4 Six tests and two comparison scales

Set

$$C = L - L', \qquad I = L - \beta, \qquad T = \operatorname{Tr} a, \qquad a_0 = 1, \quad a_1 = \beta^4 + T^4.$$

The six field tests are $C, I, f(H), f(U), \sigma f(H - P), \sigma f(V_1)$. Write $L = L_0^{\mathrm{prim}} + \rho L_1^{\mathrm{prim}}$, with rational components. In the quadratic source retain the rational component I_0 of I, with

$$I_0 - L_0^{\mathrm{prim}} + 2X^2 + B = 0.$$

Thus $\beta = L_0^{\mathrm{prim}} - I_0$ is affine in the retained coordinates, and $I = I_0 + \rho L_1^{\mathrm{prim}}$. The fixed shift B occurs only in this one equation. Retain the four reciprocal squares and C^2, I^2 as quadratic outputs. Their squares will occur directly in quadratic factor equations. The two fields in (9) supply two reciprocal squares directly. Three scalar outputs compute the comparison scale without taking any root or making any division:

$$s_h = \beta^2 + T^2, \quad p_h = \beta T, \quad a_1 = s_h^2 - 2p_h^2 = \beta^4 + T^4.$$

Retain a rational λ. For $a \in \{a_0, a_1\}$ put $t = a\lambda$, $\delta = t - 1$, and use scalar coordinates r, q, c, k satisfying

$$\begin{aligned} r = \delta^2, \qquad c = qr, \qquad k = cr, \\ k^2 - 30kc + 273c^2 - 820cq + 576q^2 = kr. \end{aligned} \tag{11}$$

Define, without storing another coordinate,

$$\begin{aligned} F &= (r-1)(r-4)(r-9)(r-16) \\ &= r^4 - 30r^3 + 273r^2 - 820r + 576 = \prod_{\theta \in \Theta_8} (t - \theta). \end{aligned}$$

The displayed equations imply $q^2F = qr^3$ and $k = qr^2$. The reciprocal-factor equations below exclude $q = 0$, so this clearing recovers $qF = r^3$ without a coordinate for F. The two field factors will have the values

$$\begin{aligned} T_1 &= 1 + qC^2 + q^2 f(H)^2 + q^4 f(U)^4, \\ T_2 &= 1 + qI^2 + q^2\sigma f(H - P)^2 + q^4\sigma f(V_1)^4, \qquad W = T_1T_2 = w_0 + w_1\rho. \end{aligned} \tag{12}$$

Their literal quadratic specification uses three scalar reciprocal weights s_2, s_3, s_4:

$$s_4 = s_2^2, \qquad qs_4 = s_3, \qquad qs_3 = s_2. \tag{13}$$

The two quadratic field equations are

$$\begin{aligned} s_4T_1 &= s_4 + s_3C^2 + s_2 f(H)^2 + (f(U)^2)^2, \\ s_4T_2 &= s_4 + s_3I^2 + s_2\sigma f(H - P)^2 + (\sigma f(V_1)^2)^2. \end{aligned} \tag{14}$$

Each fourth power is evaluated directly as a quadratic polynomial in the two components of its stored square. The lemma below proves $s_2 \neq 0$ and hence $s_j = q^{-j}$; multiplying (14) by q^4 then gives (12) exactly. Finally retain scalars τ, b with

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad kb = \tau^2. \tag{15}$$

The normalized pole uses seven further rational coordinates e, u, p, v, w, p_2, w_2 and five quadratic equations

$$\begin{aligned} p_2 = p^2, \qquad 17bp_2 = 289 + 34e - 3e^2, \\ e(17 - b) = 17uv, \qquad w_2 = w^2, \qquad 17bw_2 = u^2 + 34uv - 867v^2. \end{aligned} \tag{16}$$

For $b \neq 0$ these give (1) under

$$e_0 = e/17, \quad p_0 = bp, \quad s_0 = bu, \quad q_0 = b^2uw.$$

Conversely a witness with $s_0 \neq 0$ gives the displayed normalized coordinates by these formulas and $v = e(17 - b)/(17u)$. This explains both the pole obstruction and its use in completeness.

Lemma 4.1. *Every rational solution of the quadratic source has $\delta Fqk \neq 0$ in both detectors. Consequently*

$$q = \frac{\delta^6}{F}, \qquad k = \frac{\delta^{10}}{F} = \frac{1}{R}, \qquad R = \frac{F}{\delta^{10}}.$$

Proof. Suppose $s_2 = 0$. Equations (13) give $s_4 = s_3 = 0$, and (14) gives $f(U)^4 = \sigma f(V_1)^4 = 0$. Since K is a field, $f(U)^2 = f(V_1)^2 = 0$. The identities in the pair-chart proof give $V^4 = Q^4 = 0$. Thus $V = Q = 0$, and $VQ = aW_A$, with $a \neq 0$, forces $W_A = 0$. The last equation of (8) would then say $0 = 1$. This contradiction proves $s_2 \neq 0$. Now $qs_3 = s_2$ gives $q \neq 0$. Substitution into $s_4 = s_2^2$ and $q^2s_4 = s_2$ gives $s_2 = q^{-2}$, and then $s_3 = q^{-3}$ and $s_4 = q^{-4}$. This justifies (12) before any valuation or formal-parameter argument. The multiplied contact equation consequently gives $qF = r^3$. If $r = 0$, then $F = 576 \neq 0$ and the latter identity gives $q = 0$, again impossible. Thus $r = \delta^2 \neq 0$. Finally $F = 0$ would imply $r = 0$. The asserted formulas and the nonvanishing of $k = qr^2$ follow. In particular $\lambda \neq 0$ because zero is a contact. □

5 Local extraction and denominator exclusions

At a prime $p > 76176$, the curve has good reduction and D is a unit. For a nonzero finite reciprocal abscissa $f(Q)$, its valuation at either place of K above p is even. If $v(x) < 0$, the curve equation gives $2v(y) = 3v(x)$; if $v(x) > 0$, it gives $2v(y) = v(x)$. The unit case is immediate. Zero reciprocal abscissas are interpreted as having infinite valuation. Lemma 3.1 identifies both stored reciprocals at every solution, including the exceptional leading-coefficient cases of the quadratic chart. Their even valuations are therefore available before any formal-parameter quotient is interpreted.

Let $p \mid M(t)$, put $\nu = v_p(R)$, and normalize valuations by $v(p) = 1$. Then ν is positive and odd, δ is a unit, and $v(q) = -\nu$. For each factor in (12), the three nonconstant finite valuations belong respectively to

$$1 + 2\mathbb{Z}, \qquad 2 + 4\mathbb{Z}, \qquad 4 + 8\mathbb{Z}.$$

Thus all term valuations, including zero for the constant term, are distinct. Each factor is nonzero and has valuation at most zero; valuation zero implies residue one. A zero test merely removes a term.

Proposition 5.1. *At every contact prime, there is one place at which all six tests have positive valuation. The four reciprocal-abscissa tests have valuation at least two.*

Proof. The preceding argument gives $W \neq 0$. Equation (15) then implies $w_0 \neq 0$, since $w_0 = 0$ would imply $w_1 = 0$. Hence

$$\tau = \frac{\mathrm{N}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0, \qquad b = R\tau^2 \neq 0.$$

Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality gives $v_p(b) \leq \nu + 2\ell$. This valuation is odd, so the pole obstruction forces $0 < v_p(b) \leq \nu + 2\ell$, hence $\ell > -\nu/2$. At a place attaining ℓ, each factor valuation is at least ℓ. Each of its terms therefore has valuation greater than $-\nu/2$. For the three columns this gives respectively

$$v(C), v(I) > \nu/4, \qquad v(f(H)), v(\sigma f(H - P)) > 3\nu/4, \qquad v(f(U)), v(\sigma f(V_1)) > 7\nu/8.$$

Integer rounding and evenness give the assertion. □

The finite H chart may contain torsion translates. Before interpreting either ratio, exclude

$$E[2](K) \ \cup \ (P + E[2](K)) \ \cup \ (-P + E[2](K)).$$

Every permitted point in this finite set has reciprocal abscissa a unit above $p > 76176$. The torsion translation formulas, together with

$$\mathrm{N}(x_P) = 41^2/64, \quad \mathrm{N}(x_P - d) = 17^2/64, \quad \mathrm{N}(x_P + d) = 63^2/64,$$

give this directly. If H belonged to the set, Proposition 5.1 would force both contact radicals to equal one. Lemmas 2.1 and 2.2, and $h(a_1) \leq h(a_1\lambda) + h(\lambda)$, would then give $h(1 : \beta : T) \leq \frac{8}{3}C_0 + C_1$. Since $\beta \geq B$, this is impossible for every sufficiently large fixed B.

Consequently $h_* \notin \{-1, 0, 1\}$. Both reference points are nontorsion, so $ST_H \neq 0$ and $z(H), z(H - P)$ are defined and nonzero. Equations (10) now have their stated quotient interpretation. This exclusion does not use a logarithmic ratio and therefore precedes its use.

6 Soundness for all sufficiently large shifts

Write $U = [u]P$, so $V_1 = [u-n]P$. At a contact place supplied by Proposition 5.1, H, U lie in the primary formal group and $H - P, V_1$ in the conjugate formal group. The formal parameter satisfies

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) \equiv 1 \pmod{p^{4v(z(Q))}}. \tag{17}$$

This is the constant-term case of the proved formal-logarithm estimate in [6]. The unit is defined to be one at the identity.

The integral group identities

$$[m]U = [2u]H, \qquad [m-2]V_1 = [2(u-n)](H-P)$$

kill the torsion component of H. Their full logarithmic ratios are $r = u/h_*$ and $r' = (u-n)/(h_*-1)$. At least one of h_*, h_*-1 is a p-adic unit, so at least one full ratio is integral. The units in (17) preserve the corresponding approximate valuation, and the positive-order test $C = L - L'$ transfers integrality to the other approximate ratio and hence to both full ratios. Only after this step does (17) give absolute congruences $L \equiv r$ and $L' \equiv r' \pmod{p}$. Therefore

$$n = h_* r - (h_*-1)r' \equiv L \equiv \beta \pmod{p}.$$

The argument also covers identity numerator points and half-integral h_*, since p is odd. In particular β is integral at p.

If $\beta \neq n$, let N_β be the positive absolute reduced numerator of $\beta - n$. Both $M(\lambda)$ and $M(a_1\lambda)$ divide N_β. Lemma 2.1 gives

$$h(a_1) \le h(\lambda) + h(a_1\lambda) \le \frac{2}{3}\log N_\beta + 2C_0.$$

Put $H_2 = h(1 : \beta : T)$. Lemma 2.2 implies

$$H_2 \le \frac{8}{9}\log N_\beta + C_2, \qquad C_2 = \frac{8}{3}C_0 + C_1.$$

For a reduced fraction $\beta = a/b$, $|a - nb| \le (1+|n|)\max(|a|, b)$. Since $h(\beta) \le H_2$,

$$\log N_\beta \le H_2 + \log(1+|n|), \qquad H_2 \le 8\log(1+|n|) + 9C_2.$$

The trace-height bound (3) now yields

$$n^2 \le 8A\log(1+|n|) + 9AC_2 + J.$$

Its right side grows logarithmically, so $|n|$ and then H_2 are uniformly bounded. Every constant here is independent of B. Since $\beta \ge B$, choosing one sufficiently large odd shift excludes $\beta \neq n$. Thus $\beta = n \in \mathbb{Z}$, and consequently $X \in \mathbb{Z}$. Together with the earlier exceptional-point exclusion this proves soundness for every odd B beyond one fixed threshold.

7 Simultaneous completeness

Fix $X \in \mathbb{Z}$ and an odd $B \ge 3$. Put $n = \beta$ and $A = [n]P$; this fixes T and both positive rational scales. Choose a nonzero reference integer divisible by 64 and compact real and finite-place neighborhoods for the points

$$H = [h]P, \quad U = [hn]P, \quad H - P = [h-1]P, \quad V_1 = [(h-1)n]P$$

which avoid the finitely many chart denominators. In these neighborhoods every rational test has a fixed lower valuation bound at the finitely many preparation primes. All these choices precede prime selection.

Choose a fixed $D_1 > 0$ clearing $D_1^2\theta/a$ for both scales and all contacts, and sufficiently divisible at the preparation primes. Apply the retained prime-pattern input to the distinct coefficients $-D_1^2\theta/a$, with repetitions removed, and write

$$\lambda = \frac{u}{D_1^2 w^2}, \qquad L_1 = D_1^2 w^2.$$

Its prime values are $u - (\theta/a)L_1$. The zero contact makes u one of the selected primes. The supplied conjugate reduction-order conditions and the compatible congruence progression in [2] give h for which H, U are formal at one place above each selected prime and $H - P, V_1$ are formal at the conjugate place. The fixed congruence and real neighborhoods are retained. Only natural depth one is requested at either branch.

Both full logarithmic ratios equal n. Hence the four reciprocal tests have valuation at least two and C, I have valuation at least four at the chosen place. Since $v_p(q) = -1$, every nonconstant term of both factors in (12) has positive valuation there. At the other place each factor is either a unit of residue one or has negative valuation. Thus $\mathrm{Tr}(W^{-1})$ has residue one or two and is a unit. The pole input has valuation one at every selected contact prime.

For arbitrary positive rational a the exact identity

$$R = \frac{a^8 \prod_{\theta\in\Theta_8}(u - (\theta/a)L_1)}{L_1^8(a\lambda - 1)^{10}} \tag{18}$$

has only square factors apart from the selected prime numerators. In particular no square condition on a is needed. At every other prime $v_p(R)$ is even, and the reciprocal trace changes valuations by an even number. These facts give a square reduced denominator for b.

At a preparation prime, increasing the fixed denominator divisibility makes $v_p(t) \ll 0$. Then

$$v_p(q) = v_p(R) = -2v_p(t), \qquad R = \delta^{-2} \prod_{\theta\in\Theta_8} \left(1 + \frac{1-\theta}{\delta}\right).$$

The product is arbitrarily close to one and is a local square at $2, 3, 17$. All weighted test terms are small, so $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two. Sufficient divisibility therefore gives the three local squareclasses and valuation depths in (2).

On the fixed compact real neighborhoods, as the lower bound on λ tends to infinity, $R \sim q \sim t^{-2}$, $W \to 1$, and $b \sim 1/(4t^2)$. Both real embeddings of every factor are positive for sufficiently large λ. Hence $0 < b < 17$ and every required denominator is nonzero. These bounds are uniform on the chosen compact sets, so they can be fixed before applying the prime-pattern theorem. Each prepared b has a nonzero pole witness, which converts to (16). All quadratic coordinates can now be filled by their stated rational values. This proves completeness for every integer input and every odd $B \geq 3$.

8 The polynomial and its exact certificate

Each field value represents two rational coordinates. For the root abscissa a, the four reciprocal squares and C^2, the stored second coordinate is one half of the ρ coefficient. Other stored field values use their ordinary coefficients. These rational changes of coordinates are invertible; they are fixed in the coefficient tables. The six component equations of (5) are first made primitive, then replaced by the nonsingular rational row combinations recorded in the coefficient certificate. Their row-change determinant is $-1/362963666515968$. Each resulting row, and each remaining component equation, is cleared to a primitive integer quadratic, with positive

first nonzero coefficient in the prescribed monomial order. These operations preserve their common zero set. Their sum of squares defines P_B.

The coordinate count is

Coordinates	Number
Input, its rational comparison component, and the common slope	3
Root chart for A	6
Common chart for $H, H - P$	8
Pair chart for U, V_1	10
The two ratios	4
Four reciprocal squares and two comparison squares	12
Three scalar comparison-scale outputs	3
Two detector blocks, including their pole coordinates	44
The second argument $a_1\lambda$	1
Total	91

Each detector uses two contact outputs r, c, four weight coordinates, four factor components, two product components, the three scalars τ, k, b, and seven pole coordinates: twenty-two in all. There are 79 primitive quadratic equations, so over $\mathbb{Q}$ their sum of squares vanishes exactly when they all vanish.

The complete specification is in `computation/quartic/family_abc91_v19`. The builder and independent checker reconstruct all 79 rows and every coefficient of the full quartic. The programs are `build_reciprocal_factors_v19.py` and `check_reciprocal_factors_v19.py`. There are 927 terms when powers of B are stored separately, and 926 ordinary monomials for every positive integer B. The fourth power of the rational component of the root coordinate h has coefficient one, proving degree four and content one. Only the constant coefficient contains two powers of B; it is $B^2 + 797683$. The remaining three coefficients involving the parameter are $4B, -2B, 2B$, at the monomials X^2, L_0^{prim}, and I_0, respectively. Every other coefficient is independent of B, and the largest absolute value among them is 4153456. Therefore

$$H(P_B) = \max(4153456, B^2 + 797683) = B^2 + 797683 \quad \text{for odd } B \geq 1833.$$

No coefficient cancellation changes the stated support for positive integer B.

The independent checker imports no producer. It reconstructs all changed reciprocal-weight and factor rows directly from their field formulas, and compares all 65 unmodified rows with their pinned source. It reconstructs the two additional reciprocal rows and expands the entire sum of 79 quadratic squares. The inherited common-point checker also verifies the nonsingularity of the rational row change. The separate standard-library program `computation/elliptic_constraints/check_fixed_difference_chart.py` verifies twelve general polynomial identities, including elimination without division, the two curve equations, translation by P, and the exceptional leading-coefficient cases. The coefficient checker also verifies the nonzero exceptional constant for the specified point P. These are finite algebraic certificates. The arithmetic assumptions and the ineffective sufficient shift remain those stated in Section 1.

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